

# LINJIE QIU

Ph.D. Student in The Hong Kong University of Science and Technology

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## Research Interests

Interests: My research focuses on **Agentic AI**, **Industrial AI**, and **Human-AI Collaboration**. I am interested in developing production-ready agents for real-world industrial workflows, with an emphasis on efficient inference, memory, coordination, and tool use, as well as secure, private, auditable, and human-supervisable deployment.

Question: How can we build **production-ready industrial agents** that integrate fragmented engineering data and legacy tools to improve productivity while reducing computational cost and end-to-end latency, and ensuring verifiable operation?

## Education

- 2026 – Present    **The Hong Kong University of Science and Technology** – Guangzhou, China  
Ph.D. student in Internet of Things, Information Hub
- Committee: Xudong Wang (Supervisor) and Mingming Fan (Co-supervisor)
  - Fully funded by Doctoral Postgraduate Studentship Award
- 2024 – 2026    **The Hong Kong University of Science and Technology** – Guangzhou, China  
M.Phil. in Computational Media and Art, Information Hub
- Thesis: Modeling Action–Response Coupling for Sustainable Virtual Reality Interaction
  - Committee: Mingming Fan (Supervisor), Hai-Ning Liang (Chair), Zeyu Wang, Li Chen
  - Fully funded by Red Bird M.Phil. Postgraduate Studentship Award
- 2020 – 2024    **Xiamen University** – Xiamen, China  
B.Eng. in Digital Media Technology  
Department of Software Engineering, School of Informatics

## Publications

### FULL CONFERENCE AND JOURNAL PAPERS (P)

In the field of computer science, top-tier conferences, in addition to journals, are venues for publishing highly selective, double-blind and peer-reviewed publications.

\* Equal Contribution.

- 2025 [P.8]    **Visual Generative Artificial Intelligence (GenAI) in Mixed Reality (MR).**  
Duotun Wang, Linjie Qiu, LinLin Cai, Yue Jiang, Mingming Fan, Zeyu Wang.  
*International Journal of Human–Computer Interaction. (IJHCI 2026)*
- 2026 [P.7]    **Spatula: Exploring On-Demand In-Situ Interfaces and Interaction for Attribute Control**  
Boyu Li, Linjie Qiu, Lin-Ping Yuan, Duotun Wang, Yue Jiang, Zeyu Wang, Hongbo Fu.  
*ACM Symposium on User Interface Software and Technology (UIST 2026)*
- 2026 [P.6]    **Follow the Signs or the Crowd? Effects of Environmental Load and Crowd Dynamics in VR Evacuation.**  
Zheng Wei, Jingchen Gao, Zhe Yan, Yihan Liu, Linjie Qiu, Xian Xu, Yun Huang, Huamin Qu.  
*IEEE Transactions on Visualization and Computer Graphics. (TVCG & IEEE VR 2026)*

- 2026 [P.5] **Direct vs. Score-based Selection: Understanding the Heisenberg Effect in Target Acquisition across Input Modalities in Virtual Reality.**  
Linjie Qiu, Duotun Wang, Boyu Li, Jiawei Li, Yulin Shen, Zeyu Wang, Mingming Fan.  
*IEEE Transactions on Visualization and Computer Graphics. (TVCG & IEEE VR 2026)*
- 2025 [P.4] **MultiUX: A Human-AI Collaborative Tool to Facilitate Multiple Usability Test Video Analyses.**  
 Luyao Shen, Qing Shi, Emily Kuang, Linjie Qiu, Shixu Zhou, Pan Hui, Mingming Fan.  
*International Journal of Human-Computer Interaction. (IJHCI 2025)*
- 2025 [P.3] **DesignMemo: Integrating Discussion Context into Online Collaboration with Enhanced Design Rationale Tracking.**  
 Boyu Li, Linjie Qiu, Duotun Wang, Qianxi Liu, Ryo Suzuki, Mingming Fan, Zeyu Wang.  
*Proceedings of the ACM on Human-Computer Interaction. (CSCW 2025)*
- 2025 [P.2] **ExplorAR: Assisting Older Adults to Learn Smartphone Apps through AR-powered Trial-and-Error with Interactive Guidance.**  
 Jiawei Li, Linjie Qiu, Zhiqing Wu, Qiongyan Chen, Ziyang Wang, Mingming Fan.  
*ACM Multimedia 2025. (ACM MM 2025)*
- 2025 [P.1] **FocalSelect: Improving Occluded Objects Acquisition with Heuristic Selection and Disambiguation in Virtual Reality.**  
 Duotun Wang\*, Linjie Qiu\*, Boyu Li, Qianxi Liu, Xiaoying Wei, Jianhao Chen, Zeyu Wang, Mingming Fan. *IEEE Transactions on Visualization and Computer Graphics. (TVCG & IEEE VR 2025)*

## MANUSCRIPTS (M)

- 2026 [M.1] Anonymous authors. (The First author) *Ready for submission to IEEE VR*  
**Combating Rotation-Induced VR Sickness through Body-Coupled Turning**

## Honors, Awards, and Grants

- 2026 – 2030 **Doctoral Postgraduate Studentship Award (¥ 720,000)**, HKUST(GZ)
- 2026 **Red Bird M.Phil. Research Travel Grant (¥ 10,000)**, HKUST(GZ)
- 2025 **Diversity, Equity, Inclusion, and Accessibility Grant (¥ 1,000,000)**, IEEE VR
- 2024 – 2026 **Red Bird M.Phil. Postgraduate Studentship Award (¥ 240,000)**, HKUST(GZ)

## Professional Experience

Fall 2023 – **The Hong Kong University of Science and Technology**, Guangzhou, China  
Present Research Intern (Fall 2023 – Fall 2024) and M.Phil. Researcher (Fall 2024 – Fall 2026)  
Advisor: [Mingming Fan](#).

- Designed and evaluated occlusion-aware object acquisition techniques for distant and hidden targets, combining heuristic target prediction with interactive disambiguation to improve selection reliability. [P.1]
- Modeled confirmation-induced perturbations in VR selection by comparing direct and score-based mechanisms across hand and controller input, with analyses of error types, target width, spacing, and action-induced ray deviation. [P.5]
- Implemented a body-coupled virtual turning system that maps subtle bimanual forearm rotations to virtual yaw rotation for combating the VR sickness. [M.2]

## Invited Talks

Dec. 2025 **Intelligent Interface: Supporting User Intent in Spatial and Collaborative Contexts**  
University of Illinois Urbana-Champaign, Remote.

## Academic Services

### Reviewer (Selected)

I have reviewed 7 papers (including short papers) in top-tier human-computer interaction, extended reality, visualization and computer graphics venues.

- 2026 SIGCHI Conference on Human Factors in Computing Systems (CHI)  
2026 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)

## Skills

Programming Python, C#, C++, C, R, HTML/CSS/JavaScript, Kotlin  
Frameworks PyTorch, Flask, Django, Vuforia  
Creative Tools Unity, Maya, R Studio, Android Studio, Meta Quest, Microsoft HoloLens  
Language Chinese (Native), English (Professional Working Proficiency: TOEFL iBT 102)

## Referees

- R.1 **Prof. Xudong Wang** – [xudongwang@hkust-gz.edu.cn](mailto:xudongwang@hkust-gz.edu.cn)  
Professor, The Hong Kong University of Science and Technology (Guangzhou), CN  
IEEE Fellow & Acting Dean, the College of Future Technology
- R.2 **Prof. Mingming Fan** – [mingmingfan@ust.hk](mailto:mingmingfan@ust.hk)  
Associate Professor, The Hong Kong University of Science and Technology, CN  
Acting Head, Red Bird MPhil Program
- R.3 **Prof. Zeyu Wang** – [zeyuwang@ust.hk](mailto:zeyuwang@ust.hk)  
Assistant Professor, The Hong Kong University of Science and Technology, CN
- R.4 **Prof. Yue Jiang** – [yue.jiang@utah.edu](mailto:yue.jiang@utah.edu)  
Assistant Professor, University of Utah, USA